

Ch 1: Real Numbers

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Find the HCF and LCM of 96 and 404 by prime factorisation method. (PYQ 2018)

Q2. Find the HCF and LCM of 336 and 54 using prime factorisation.

Q3. Given $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$. (PYQ 2018)

Q4. The HCF of two numbers is 9 and their LCM is 2016. If one number is 54, find the other. (PYQ 2020)

Q5. If $\text{HCF}(6, a) = 2$ and $\text{LCM}(6, a) = 60$, find a .

Q6. The LCM of two numbers is 14 times their HCF. The sum of LCM and HCF is 600. If one number is 280, find the other.

Q7. Find the LCM and HCF of 12, 15, and 21 by the prime factorisation method.

Q8. Find the HCF of 144 and 180. If HCF is expressed as $13m - 3$, find the value of m .

Q9. Check whether 6^n can end with the digit 0 for any natural number n . (PYQ 2018, 2020)

Q10. Check whether 4^n can end with the digit 0 for any natural number n .

Q11. Show that 9^n cannot end with digit 0 for any natural number n .

Q12. Show that 5^n cannot end with digit 2 for any natural number n .

Q13. Explain why $7 \times 11 \times 13 + 13$ is a composite number. (PYQ 2018)

Q14. Explain why $3 \times 5 \times 7 + 7$ is a composite number.

Q15. Explain why $2 \times 3 \times 5 \times 7 \times 11 \times 13 + 13$ is a composite number.

Q16. Show that $\sqrt{3}$ is irrational. (PYQ)

Q17. Show that $\sqrt{5}$ is irrational. (PYQ 2018)

Q18. Show that $\sqrt{2}$ is irrational. (PYQ 2021, 2022)

Q19. Show that $2 + \sqrt{3}$ is irrational. (PYQ 2020)

Q20. Show that $3\sqrt{2}$ is irrational.

Q21. Show that $5 - \sqrt{3}$ is irrational.

Q22. Show that $2 - \sqrt{5}$ is irrational.

Q23. Show that $1/\sqrt{2}$ is irrational.

Q24. Three bells toll at intervals of 9, 12, and 15 minutes. If they start tolling together, after what time will they toll together again?

Q25. Find the largest number that divides 70 and 125 leaving remainders 5 and 8 respectively. *(PYQ 2018)*

Q26. Two tankers contain 850 litres and 680 litres of petrol. Find the maximum capacity of a container that can measure the petrol of either tanker in exact number of times.

Q27. If two positive integers p and q are expressed as $p = ab^2$ and $q = a^3b$ where a and b are prime numbers, find HCF (p, q) and LCM (p, q). *(PYQ 2019)*

Q28. If two positive integers m and n are expressed as $m = p^2q^3$ and $n = p^3q^2$, where p and q are prime numbers, find HCF and LCM of m and n .

Q29. Find the smallest number which when divided by 35, 56, and 91 leaves remainder 7 in each case.

Q30. Two numbers are in the ratio 15:11. Their HCF is 13. Find the numbers and their LCM.

SECTION B - 3 MARKS

Q1. Prove that $\sqrt{2}$ is irrational. (PYQ 2022, 2021, 2020, 2019 — most repeated)

Q2. Prove that $\sqrt{3}$ is irrational. (PYQ 2020)

Q3. Prove that $\sqrt{5}$ is irrational. (PYQ 2018)

Q4. Prove that $3 + 2\sqrt{5}$ is irrational. (PYQ 2020, 2019)

Q5. Prove that $2 + \sqrt{3}$ is irrational. (PYQ 2020)

Q6. Prove that $5 - 2\sqrt{3}$ is irrational. (PYQ SQP 2021-22)

Q7. Prove that $3 - \sqrt{5}$ is irrational.

Q8. Prove that $5 + 3\sqrt{2}$ is irrational.

Q9. Prove that $7 - 2\sqrt{3}$ is irrational.

Q10. Prove that $(2 + \sqrt{3})/5$ is irrational, given that $\sqrt{3}$ is irrational.

Q11. Prove that $\sqrt{2} + \sqrt{3}$ is irrational. (PYQ HOTS)

Q12. Prove that $\sqrt{3} + \sqrt{5}$ is irrational.

Q13. Prove that if p is a prime number, then $\sqrt{p}$ is irrational. (PYQ 2020)

Q14. Find the HCF and LCM of 510 and 92 by prime factorisation. Verify that $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$.

Q15. Find the HCF and LCM of 26, 65, and 117 using prime factorisation.

Q16. Find the LCM and HCF of 6, 72, and 120 using prime factorisation.

Q17. Find HCF and LCM of 90 and 120. Verify $\text{HCF} \times \text{LCM} =$ product of the two numbers. (PYQ 2019)

Q18. The HCF of 65 and 117 is expressible in the form $65m - 117$. Find the value of m . (PYQ 2019)

Q19. Show that any positive odd integer is of the form $6q + 1$, $6q + 3$, or $6q + 5$ where q is some integer. (PYQ 2020)

Q20. Show that any positive odd integer is of the form $4q + 1$ or $4q + 3$ where q is a non-negative integer.

Q21. Show that the square of any positive integer is of the form $3m$ or $3m + 1$.

Q22. Show that the square of any positive integer cannot be of the form $5q + 2$ or $5q + 3$.

Q23. If n is an odd positive integer, show that $n^2 - 1$ is divisible by 8.

Q24. Show that the product of three consecutive positive integers is divisible by 6. (PYQ 2013)

Q25. Show that one and only one of n , $n+2$, $n+4$ is divisible by 3.

Q26. Three sets of English, Mathematics, and Science books have to be stacked in such a way that all books are stored

topic-wise and the height of each stack is the same. There are 96 English, 240 Mathematics, and 336 Science books. Find the number of stacks of each subject.

Q27. In a morning walk three persons step off together. Their steps measure 80 cm, 85 cm, and 90 cm. What is the minimum distance each should walk so that all can cover the same distance in complete steps?

Q28. Amita, Sneha, and Manya begin to jog around a circular stadium and complete their rounds in 42, 56, and 63 minutes respectively. After how many minutes will they come together at the starting point?

Q29. Find the greatest number that will divide 445, 572, and 699 leaving remainders 4, 5, and 6 respectively.

Q30. Find the largest number which divides 245 and 1029 leaving remainder 5 in each case.

Q31. Find the largest number which divides 2053 and 967 leaving remainder 5 and 7 respectively.

Q32. Show that $5 \times 7 \times 11 \times 13 + 13$ is a composite number. Explain using Fundamental Theorem of Arithmetic.

Q33. Show that $n^2 - n$ is divisible by 2 for every positive integer n .

Q34. Prove that $n^3 - n$ is divisible by 6 for every positive integer n .

**Q35. If p is a prime and p divides a^2 , prove that p also divides a .
(PYQ HOTS)**

Q36. Find the greatest 3-digit number which is exactly divisible by 8, 10, and 12.

Q37. Find the smallest 4-digit number which is divisible by 18, 24, and 32.

Q38. Two positive integers a and b can be expressed as $a = x^3y^2$ and $b = xy^3$ where x and y are prime numbers. Find LCM (a, b) and HCF (a, b).

Q39. Prove that the product of two consecutive positive integers is divisible by 2.

Q40. Using Fundamental Theorem of Arithmetic, find HCF and LCM of 420 and 130.

Q41. Show that $\sqrt{p} + \sqrt{q}$ is irrational where p and q are distinct primes.

Q42. Prove that $3 + 2\sqrt{7}$ is irrational given that $\sqrt{7}$ is irrational.

SECTION C - 5 MARKS

Q1. Prove that $\sqrt{2}$ is irrational. Hence show that $3 + 2\sqrt{2}$ is also irrational. (PYQ 2022 — *most repeated*)

Q2. Prove that $\sqrt{3}$ is irrational. Hence show that $5 + 2\sqrt{3}$ is irrational. (PYQ 2020)

Q3. Prove that $\sqrt{5}$ is irrational. Hence show that $2 - \sqrt{5}$ is irrational. (PYQ 2018)

Q4. Prove that $\sqrt{2}$ is irrational. Hence show that $(3 - \sqrt{2})/5$ is irrational.

Q5. Prove that $3 + 2\sqrt{5}$ is irrational given that $\sqrt{5}$ is irrational. Hence check if $(3 + 2\sqrt{5})(3 - 2\sqrt{5})$ is rational or irrational with justification. (PYQ 2020)

Q6. Find HCF and LCM of 404 and 96 by prime factorisation. Verify that $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$. (PYQ 2018)

Q7. Prove that if a prime p divides a^2 , then p also divides a where a is a positive integer. Using this, prove that $\sqrt{2}$ is irrational. (PYQ HOTS)

Q8. Show that any positive odd integer can be expressed in the form $4q + 1$ or $4q + 3$. Using this, prove that the product of two consecutive odd positive integers is always of the form $4k + 3$.

Q9. Prove that $\sqrt{2} + \sqrt{3}$ is irrational. (PYQ HOTS)

Q10. Find HCF and LCM of 6, 72, and 120 using prime factorisation. Verify that $\text{LCM} \times \text{HCF}$ is not equal to product of all three numbers and explain why.

Q11. A circular field has circumference of 360 km. Two cyclists Sumeet and John start together from the same point. Sumeet cycles at 12 km/hr and John at 15 km/hr. After how many hours will they meet again at the starting point? Also find the number of rounds each will have completed.

Q12. Three alarm clocks ring at intervals of 4, 12, and 20 minutes respectively. If they start ringing together, after how many minutes will they ring together again? Also find the number of times each clock rings before they all ring together again.

Q13. Prove that $\sqrt{5}$ is irrational. Hence show that $(2\sqrt{5} + 3)$ is irrational. Also check if $(2\sqrt{5} + 3)(2\sqrt{5} - 3)$ is rational or irrational.

Q14. Find the greatest number that will divide 443, 524, and 605 leaving remainders 5, 4, and 3 respectively. Also find the HCF and show how it relates to the result.

Q15. Find the smallest number which when increased by 17 is exactly divisible by both 468 and 520.

Q16. Prove that $n^2 - n$ is divisible by 2 for every positive integer n . Hence also prove that $n^3 - n$ is divisible by 6.

Q17. Two positive integers a and b can be expressed as $a = x^3y^2$ and $b = xy^3$ where x and y are prime numbers. Find LCM (a, b), HCF (a, b), and verify that $\text{LCM} \times \text{HCF} = a \times b$.

Q18. The length, breadth, and height of a room are 8 m 25 cm, 6 m 75 cm, and 4 m 50 cm respectively. Find the length of the longest rod that can measure the dimensions of the room exactly.

Q19. Prove that $\sqrt{3} + \sqrt{5}$ is irrational. Hence state whether $2(\sqrt{3} + \sqrt{5})$ is rational or irrational.

Q20. Show that the square of any positive integer is of the form $5m$, $5m+1$, or $5m+4$ and hence prove that it can never be of the form $5m+2$ or $5m+3$.